

Theory and the future of land-climate science

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Climate over land—where humans live and the majority of food is produced—is changing rapidly, driving severe impacts through extreme heat, wildfires, drought and flooding. Our ability to monitor and model this changing climate is being transformed through new observational systems and increasingly complex Earth system models. But fundamental understanding of the processes governing land climate has not kept pace, weakening our ability to interpret and utilize data from these advanced tools. Here we argue that for land-climate science to accelerate forwards, an alternative approach is needed. We advocate a parallel scientific effort, one emphasizing robust theories, that aims to inspire current and future land-climate scientists to better comprehend the processes governing land climate, its variability and extremes and its sensitivity to global warming. Such an effort, we believe, is essential to better understand the risks people face, where they live, in an era of climate change.

Knowledge of some aspects of continental climate and their responses to global warming are well established. For example, we broadly understand why land warms more rapidly than oceans¹ (Fig. 1), the intensification of extreme precipitation in a warmer atmosphere² and how surface run-off is influenced by loss of snowpack³. However, knowledge of many other aspects of land climate is underdeveloped. The ‘wet get wetter, dry get drier’ paradigm predicts an amplification of wet/dry contrasts as climate warms⁴. But this paradigm does not generally apply to land regions⁵; neither does the poleward expansion of the Hadley cells⁶. Adding to this list is uncertainty over how evapotranspiration (ET) and soil moisture^{7,8}—both critical for humans and ecosystems—will be altered by a changing climate. Knowledge of numerous other facets of land climate is similarly unsettled, from basic questions of what governs its mean state, variability and extremes to how these facets might change with warming. Why are simulated land temperature changes more uncertain and more diverse, across space and climate models,

compared with ocean regions (Fig. 1a,b)? Why are the tropical rain belts broader and more mobile over land⁹? And how will land humidity evolve as climate warms¹⁰? Long-standing challenges in simulating land climate—including the diurnal cycle of convection¹¹—further highlight shortcomings in our basic understanding.

The challenge of complexity

The climate over land is a complex system shaped by an array of diverse factors, from local surface conditions, including soil moisture and plants^{12,13}, to large-scale atmospheric circulations that connect continents to oceans through the transport of water, heat and momentum^{14,15}. Many of the key processes influencing land climate are spatially heterogeneous, difficult to simulate and/or poorly observed. For example, land surface models have long-standing problems in simulating turbulent fluxes of heat and water^{16,17}, for reasons that are not well understood¹⁸. Sparse and time-limited observational records

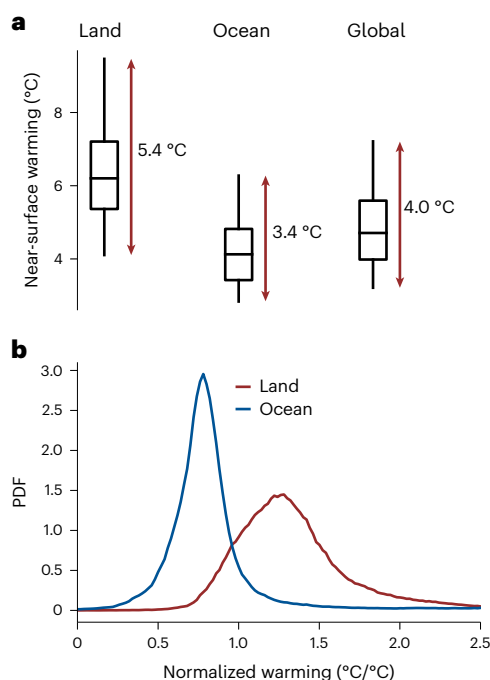


Fig. 1 | Simulated climate warming is larger and more uncertain over land. **a**, Box plots of simulated warming averaged over land (left), over ocean (centre) and globally (right) calculated using pre-industrial control and abrupt $4 \times \text{CO}_2$ simulations performed by 45 climate models participating in the Coupled Model Intercomparison Project Phase 6⁹⁶. Horizontal lines show the median model values, boxes show the interquartile ranges, and whiskers show the full model ranges. Warming for each model is computed as the time- and area-averaged near-surface temperature change between the final 20 years of the pre-industrial control simulation and years 40–59 of the abrupt $4 \times \text{CO}_2$ simulation. Uncertainty across models is indicated by the red arrows and text, with the full model range taken as a simple measure of uncertainty. **b**, Multimodel-mean probability density functions (PDFs) of area-weighted near-surface warming over land (red) and ocean (blue), normalized by the global-mean warming in each model. The same models, simulations and averaging periods are used as in panel **a**. The wider land PDF in panel **b** suggests larger differences in near-surface warming, across space and models, relative to oceans.

of important land-climate variables, including root-zone soil moisture¹⁹ and near-surface humidity²⁰, further impede efforts to advance knowledge of the land-climate system. The role of humanity presents another challenge, with large uncertainties in modelling the influences of land use and management on fluxes of carbon, energy and water in the past, present and future²¹. **Confronted with such a complex system, it can appear a daunting task to develop a deep, mechanistic, conceptual understanding of the kind we would want to read in future textbooks on land climate.** But as the field of climate science evolves, we argue that many of the most fascinating and pressing questions relate to land.

Given the complexity and importance of land climate, how can the research community accelerate progress? **In the atmospheric and ocean sciences, notable advances are being made by increasing the spatial resolution of state-of-the-art Earth system models (ESMs)**²². Unlike in the atmosphere and oceans, however, where higher resolutions allow for explicit simulation of key processes, including deep convection and mesoscale eddies, the case for transitioning to finer-resolution models to drive new conceptual breakthroughs in land-climate science is less clear-cut²³. Land climate is undoubtedly influenced by small-scale processes, so there are potential benefits to incorporating into models more sophisticated representations of, for example, hillslope hydrology²⁴, groundwater processes²⁵ and land management²⁶. However, **complexity does not equate to**

realism; absent a comprehensive understanding of these processes and how to accurately represent them in models²⁷, it is possible that such complexity obfuscates more than it clarifies¹⁶. Persistent and poorly constrained deficiencies in land surface models—highlighted by the PLUMBER project^{16–18}—suggest that model development alone, although vital, is unlikely to answer the key questions about land climate highlighted in the preceding. Similarly, machine-learning tools are increasingly being applied to climate science for developing ESMs²⁸, parameterizing surface fluxes²⁹ and constructing statistical emulators of land models³⁰. Indeed, recent successes highlight the potential of machine learning to build physical insight in the atmospheric and ocean sciences^{31,32}. **It remains to be seen, however, whether the tools of machine learning are capable of transforming scientific understanding of land climate.**

A renewed focus on theory

In this Perspective, we argue that for land-climate science to move forwards, we must step back and reassess our approach. **Our philosophy—borne in an era of explosive growth in model complexity and demanding simulation timetables, and shaped by a 2022 workshop at the University of St Andrews—is to redouble efforts to build robust physical understanding of land climate through the development of powerful new theories and refinement of existing conceptual frameworks.** Previous work exemplifies this approach, notably the development of theories and simple ‘toy’ models to understand the land boundary layer³³, land-atmosphere coupling³⁴ and moist convection over land³⁵. To anchor and inspire the next decade of research, we argue that now is the time to position this philosophy at the centre of land-climate science and **re-balance our activities such that theory, model development and observations are prioritized equally.**

Development of theory can, and should, proceed in parallel with the imperative to build progressively more sophisticated ESMs. Indeed, **the gap in climate science between theory and actionable information, particularly at regional scales, is typically filled by state-of-the-art models, which are also invaluable tools for testing and refining the theories advocated here.** However, **theories that distil conceptual understanding need to be at the core of land-climate science to enable the research community to compare proposed mechanisms, understand the competing roles of different processes in a coupled system and make predictions without running complex models.** Advances in theory can have **practical as well as conceptual benefits**, for example, making ET easier to estimate³⁶, increasing confidence in model projections (for example, of run-off³⁷) and underpinning physically based emergent constraints to narrow uncertainties in future climate change³⁸.

So what constitutes a successful theory in land-climate science?

The answer depends on the problem being considered, but we believe a successful theory should explain an emergent property of the climate system, be underpinned by robust process understanding and provide clear mechanistic insights that hold across a hierarchy of numerical model complexity. Theories should also, where possible, be predictive and quantitative (formulated as an equation or set of equations). Finally, and crucially, a successful theory should be tested against and supported by observational data. In the following, we highlight three recent advances in land-climate science that showcase the power of theory, before outlining our view on how a renewed focus on theory is needed to accelerate progress in land-climate science:

Land temperature and humidity changes constrained by tropical atmospheric dynamics

The role of convection and large-scale atmospheric dynamics in shaping tropical land temperature and humidity has been an important conceptual advance over recent decades^{1,39,40}. This framework emerged from efforts to understand why, under climate change, warming is stronger over land—the so-called land–ocean warming contrast³⁹. Early explanations of this phenomenon were based on the surface energy

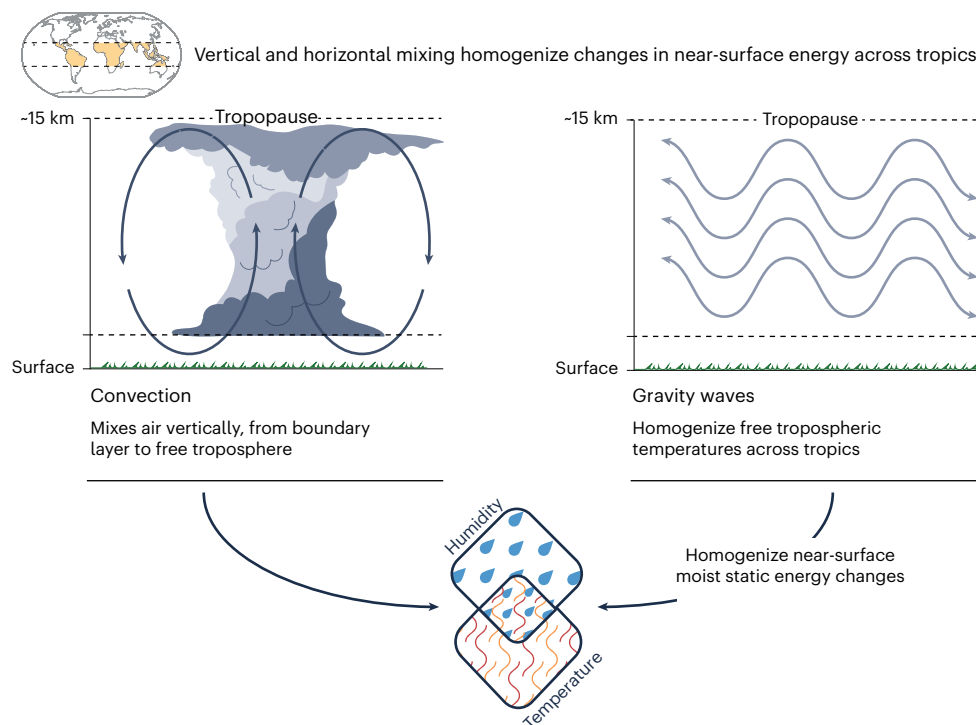


Fig. 2 | Atmospheric dynamics constrain changes in tropical land climate. Convection and gravity waves in the tropical atmosphere spatially homogenize climatic changes in near-surface moist static energy. The development of

this large-scale atmospheric constraint on tropical land climate has been an important conceptual advance over recent years. Here and in Figs. 3 and 4, the title maps highlight where the mechanism is broadly expected to be applicable.

budget⁴¹. Radiative forcing at the surface (for example, due to increases in atmospheric CO₂) are balanced in ocean regions largely by increases in evaporation, resulting in a relatively small increase in surface temperature. In land regions, however, which are often water-limited, radiative forcing is balanced primarily through increases in sensible heat and long-wave fluxes, requiring a larger increase in temperature relative to oceans. Although physically intuitive, using this argument to construct a quantitative theory for land temperature change is challenging because surface fluxes depend on multiple factors aside from temperature, including wind speed, soil moisture and the air–surface temperature disequilibrium.

An alternative framework, inspired by ref. 1, cuts through the complexity of land surfaces to reveal a strong constraint on the response of tropical land to climate change. This framework has transformed understanding of the tropical land–ocean warming contrast and has led to broader insights into large-scale atmospheric controls on near-surface temperature and humidity. In the tropical atmosphere, strong vertical coupling by convection between the boundary layer and free troposphere described by convective quasi-equilibrium⁴²—together with horizontal coupling by gravity waves above the boundary layer, resulting in weak free-tropospheric temperature gradients⁴³—imply that climatic changes in adiabatically conserved quantities such as moist static energy, a function of temperature and specific humidity near the surface, are tightly coupled between different regions and therefore approximately uniform on large scales^{44–46} (Fig. 2). This mechanism, a form of ‘downward control’ exerted by the overlying atmosphere on near-surface tropical climate, has important implications: although temperature and specific humidity individually may respond differently to climate change in different regions, for example, in tropical savannahs versus in rainforests, the combined change (encoded in the moist static energy) is more spatially homogeneous. Local processes, including soil moisture and aridity^{45,47}, are crucial for controlling how temperature versus humidity changes contribute to the change in moist static energy

imposed by the atmosphere. This physical theory underpins advances in understanding the land–ocean warming contrast^{1,48}, aridity and land relative humidity in a changing climate^{40,45,49}, and extreme heat^{46,50,51}, and establishes a simple yet quantitative framework for interpreting models, observations and the roles of local versus large-scale processes in shaping tropical land climate.

ET predicted by simple theory

ET is central to regulating the water, energy and carbon budgets of land regions⁵² and affects societies and ecosystems through its influence on hydrology and temperature variability⁵³. But ET is directly measured only at a limited number of sites⁵⁴, necessitating models of various kinds to estimate ET elsewhere. These models are typically complex, requiring numerous poorly constrained land surface parameters as inputs, and are imperfect at replicating direct measurements⁵⁵. However, a new theory to predict present-day ET in inland continental regions using minimal input data provides a conceptual advance in understanding and presents an opportunity to greatly expand the database of ET measurements across space and time³⁶. The theory is based on the concept of ‘surface flux equilibrium’ (SFE), which assumes an approximate balance between the surface moistening and heating effects on near-surface relative humidity⁵⁶. This strong coupling between the land surface and overlying atmosphere imprints, in the air properties, information about the land surface fluxes (the Bowen ratio) at daily to longer timescales and appears to dominate alternative atmospheric mechanisms that also contribute to determining the near-surface atmospheric state (for example, wind-driven moisture and heat convergence). Specifically, the SFE theory permits relatively accurate estimates of ET knowing only the net radiative flux into the surface and the near-surface temperature and specific humidity^{36,57}, the latter two of which reflect the Bowen ratio (Fig. 3). Importantly, these quantities are more widely available from weather stations than are direct ET measurements. The theory reveals an emergent simplicity in ET³⁶, despite the heterogeneity and complexity of land surfaces.

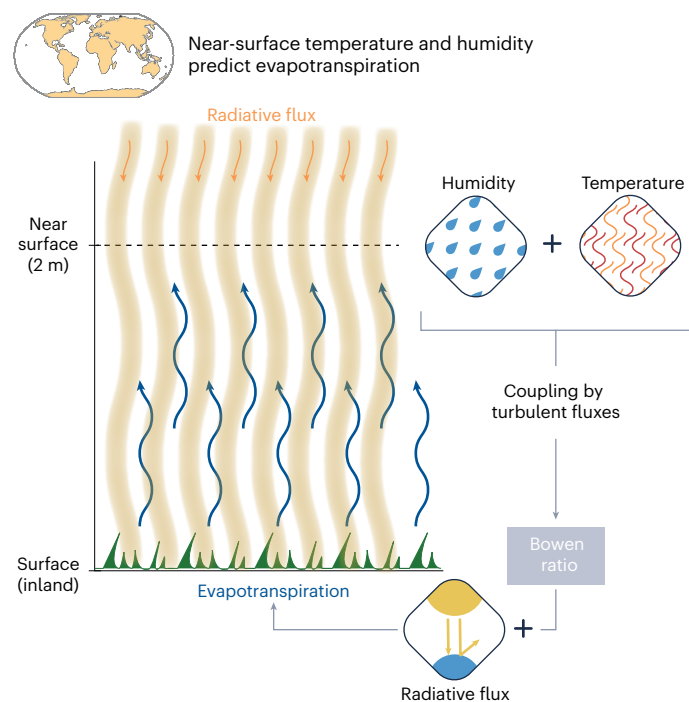


Fig. 3 | Evapotranspiration inferred from temperature and humidity measurements. Following recent theoretical developments, inland ET can be predicted as a simple function of near-surface temperature and humidity along with the net radiative flux into the surface. Note that the grey arrows represent the series of inferences used by the SFE-based theory to make estimates of ET³⁶, whereas the blue and orange arrows denote, respectively, the turbulent fluxes of heat and water coupling the surface to the near-surface air and the radiative energy fluxes.

Leaf physiology incorporated into classical run-off theories

Run-off from land supplies almost all the water used by humans. In contrast to the time-varying ET estimated by SFE and described in the preceding, long-term mean run-off and ET fluxes have long been predicted and understood using the simple theory of Budyko⁵⁸, in which the fraction of precipitation that becomes run-off decreases as the ratio of atmospheric evaporative demand to precipitation increases. Budyko quantified evaporative demand using surface net radiation only, but more comprehensive evaporative theories⁵⁹ generally also include a well-understood positive temperature dependence⁶⁰. When these more modern methods are used in the Budyko theory, they predict substantial increases in evaporative demand with global warming and systematic decreases in natural run-off⁶¹ (the component of run-off controlled by natural processes rather than by human activities), which would imply water shortages. Yet such widespread run-off declines are neither observed⁶² nor simulated by more comprehensive models⁶¹, leading to the impression of a theoretical deficiency. Reference⁶³ recently resolved this tension by incorporating the ET-reducing closure of leaf stomata by CO₂ into a revised theoretical framework (Fig. 4). The inclusion of this important and well-studied process brought the Budyko-predicted trends in natural run-off much closer to observations and state-of-the-art ESMs, and clarified our understanding of the drivers of run-off in a changing climate. Looking forwards, incorporating human activities (for example, water management) and the effects of wildfire into run-off theories is a priority for future work.

Opportunities for progress

A greater emphasis on developing theories for land climate and its changes is essential for building confidence in future projections, identifying directions for model improvement, validating in situ and

remote-sensing data and interpreting the dynamics of key processes as new models and observational systems come online. The examples highlighted in the preceding demonstrate the potential for theory to further fundamental understanding of land climate. But the next set of advances is now needed. In the following, we present three areas of land-climate science primed for theory to provide new insights.

Atmospheric circulation and land

The atmospheric circulation strongly shapes the land climate, from extreme temperatures⁶⁴ to the regional water cycle⁶⁵. However, much of our understanding of the atmospheric circulation and its sensitivity to climate change has been developed using aquaplanet models without land surfaces^{66,67}. Over recent years, focus has begun to shift towards incorporating land into conceptual frameworks for the atmospheric state and circulation^{68–70}. But numerous basic questions persist. Why is the tropical rainbelt wider over continents⁹? How can ingredients of the land surface be incorporated into modern theories for monsoons⁷¹? Why is the poleward expansion of the atmospheric circulation under global warming much weaker over land⁶? How will blocks, often the cause of extreme weather over land, change with warming⁷²? What processes control updraught velocities—and hence influence extreme precipitation—over land²? These important questions are ready to be tackled with new theories.

Water and land

Beyond a broad tendency for mean relative humidity over land to decrease with warming^{40,49,73}, basic properties of the land water cycle and its response to climate change remain unexplained. For example, what are the mechanisms determining the spatial and temporal distributions of soil moisture in the current climate⁷⁴? Why do climate models project drier surface soils in most regions⁵? Why do future trajectories for surface and column soil moisture differ⁷⁵? Detailed understanding of near-surface humidity over land is another priority¹⁰, given the strong coupling to trends in extreme temperatures^{51,76}, extreme precipitation⁷⁷ and run-off⁷⁸. The coupling between plants and water has major implications for drought and terrestrial ecosystems, yet its response to climate change is highly uncertain⁷⁹. For example, the effects of plant changes on run-off beyond the simple CO₂-stomatal dependence⁶³ are probably very large⁸⁰ but poorly understood. Finally, the phenomenon of ‘flash droughts’, whose dynamics and predictability are only beginning to be explored⁸¹, is an emerging topic where creative new theories are needed.

Carbon and land

Carbon uptake and release by terrestrial ecosystems both affects and responds to climate variability and long-term change. The field of carbon–water–climate feedbacks is already rich with examples of simple concepts, theories and emergent constraints^{82–84}, providing a way to synthesize or contrast the behaviours emerging from complex ESMs⁸⁵. The carbon-concentration and carbon-climate feedback parameters, for example, encapsulate the overall response of land carbon stocks to changes in atmospheric CO₂ and to global warming, respectively⁸⁶. This global-scale conceptual framework can be used to diagnose and compare complex simulations⁸⁷ but is also transferable to climate emulators or models of reduced complexity⁸⁸. However, similarly simple and adaptable concepts are lacking in other areas of carbon cycle research. There is, for example, large uncertainty on the extent to which tipping points at regional scales could impact some of the world’s largest carbon pools, such as permafrost carbon, the Amazon rainforest ecosystem and global forests^{89–92}. To some extent, this is because we lack theories, metrics and frameworks to explain and reconcile the contradicting results obtained from different models and approaches. However, the existing literature on dynamical systems theory is rich with concepts that may be transferable to understand potential tipping points in the carbon cycle if they can be adequately

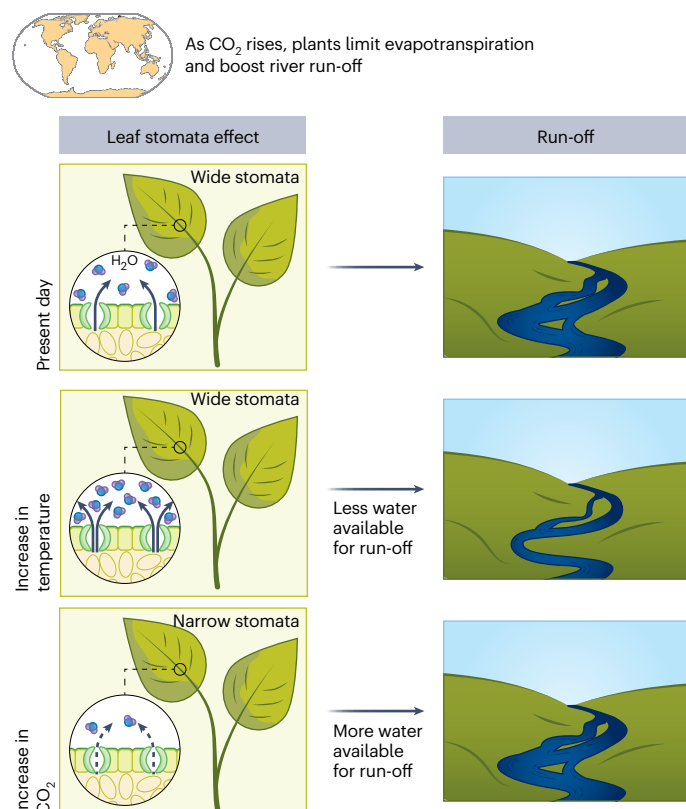


Fig. 4 | Stomatal response to increasing CO₂ boosts river run-off. The competing effects of temperature versus CO₂ on ET from leaves and on river run-off. The recent incorporation of the CO₂ effect into classical theories has clarified understanding of run-off in a changing climate.

constrained by observations, similar to what has been done to study transitions between stable system states or attractors in ecology and population dynamics^{93,94}.

Outlook

To discover, test and refine the powerful theories for land climate advocated in this Perspective, and to maximize benefits for the wider climate community, technical tools and scientific talent are needed. On the tools side, we have at our disposal a range of models spanning idealized⁹⁵ to state-of-the-art ESMs⁹⁶, alongside the emerging generation of ‘global storm resolving’ models²² and flexible, process-based hydrologic models⁹⁷. This model hierarchy is well positioned for building new understanding of land climate. However, a lack of observations presents a major challenge⁹⁸: Despite recent progress, for example, in remote sensing of surface soil moisture⁹⁹, we simply do not have long-term datasets with wide spatial coverage for many important land-climate quantities, including root-zone soil moisture and ET. Thus, to parallel the development of models and efforts to construct theories for land climate, new instrumental observations of essential land surface fluxes and reservoirs are required. Opportunities to further leverage existing observational datasets, with the goal of improving models and testing theories, should also be exploited. Beyond observational uncertainty, whenever we ground new theory in observations we also have to contend with the complicating influence of internal climate variability. Separating the forced response from internal variability at regional scales is still challenging and can harbour surprises that can influence our theories¹⁰⁰. Empirical–statistical methods to isolate the forced response, and new theory on internal variability itself, will thus need to accompany our endeavour to refine understanding of land climate and its changes with warming.

On the talent side, to tackle the important questions in land-climate science, we need to continually inspire, recruit and resource diverse cohorts of researchers from a range of primary disciplines spanning atmospheric science, hydrology, ecology, physics, mathematics, computer science and beyond. Engaging scientists from the broader climate community—those working primarily on atmospheric dynamics, for example—also has the potential to bring new ideas and drive progress in land-climate science. Through this Perspective, alongside a series of workshops and summer schools we aim to coordinate over coming years, our goal is to engage these current and future generations of researchers—as well as major funding bodies and established land-focused research initiatives—in our vision to place theory at the core of land-climate science.

State-of-the-art models, observational systems and machine learning are transforming our ability to simulate, monitor and emulate many aspects of land climate. Our scientific understanding, however, has not kept pace, and we now lack robust theories to comprehend the rich complexity being revealed by these advanced tools. Now is the time to change course and underpin models, observations and machine-learning techniques with new theories so that we maintain and advance the deep, mechanistic understanding of land climate needed to meet the challenges of an uncertain future.

Data availability

The model data used to produce Fig. 1 are provided by the World Climate Research Programme’s Working Group on Coupled Modelling and can be accessed at <https://esgf-node.llnl.gov/search/cmip6/>.

Code availability

The code used to produce Fig. 1 is available from the corresponding author on request.

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